Thermal Cracking of Fischer−Tropsch Waxes

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Thermal cracking of Fischer−Tropsch waxes at mild conditions (430−500 °C) has been reconsidered as a possible refining technology for the production of fuels and chemicals. In this study, the description of paraffin cracking, which was historically developed using material no heavier than C_{40}, was extended to C_{20}−C_{120} wax. In this carbon number range, cracking rate increased with increasing carbon length and thermal cracking resulted in a bimodal product distribution. Using the Rice−Kossiakoff mechanism as a basis, a relationship could be derived between carbon number distribution and conversion that described the observed behavior. The application of thermal cracking to fuels production was evaluated, and it was shown to be less efficient than hydrocracking, although it requires no hydrogen co-feed and has a lower gas make. Thermal cracking is better suited for the production of chemicals such as candle wax and linear α-olefins. The study also investigated the formation of deposits during wax cracking. The deposits were inorganic in nature and originated from material produced in the Fischer−Tropsch process. It was found that stripping gases (steam, H_2, and N_2) were ineffective in reducing these deposits.

Introduction

Linear paraffins and paraffin waxes from low-temperature Fischer−Tropsch (LTFT) synthesis can be used as feed for the production of both chemicals and fuels.[1] At the Sasol 1 facility in Sasolburg, South Africa, the LTFT waxes are hydrogenated and some oxidized to produce various wax-based products.[2] The Shell facility in Bintulu, Malaysia, has been designed to produce mostly transportation fuels by hydrocracking,[3] but part of the production is also directed toward the wax market.[4] A similar refinery design has been used for the Oryx facility in Ras Laffan, Qatar, with the waxes being hydrocracked to produce mainly cracker naphtha and distillates.[2]

Present commercial practice therefore suggests that waxes should either be sold as wax products, or be hydrocracked to produce transportation fuels. Yet, in recent literature this thinking is being challenged and other upgrading pathways are being explored. In addition to wax hydrocracking,[5−7] catalytic cracking of wax[8−11] and steam (thermal) cracking of the Fischer−Tropsch derived naphtha[12] are being revisited. In the past, thermal cracking of Fischer−Tropsch waxes also had been of considerable interest. It was a way to convert waxes into naphtha and distillates, as well as to produce linear α-olefins for the production of lubricating oils.[13−17] Ethylene-based routes have subsequently replaced thermal cracking of wax as a main source of linear α-olefins,[18] and hydrocracking is seen as a more efficient way to produce distillates from wax. However, process intensification[19] by tighter integration of Fischer−Tropsch synthesis with high-temperature processes such as gasification and reforming, creates possibilities for thermal upgrading in a Fischer−Tropsch refinery. Thermal cracking also has other benefits, like a low gas make and no hydrogen use.

In the present work we have reconsidered thermal cracking as a possible refining technology for Fischer−Tropsch waxes. The study focused on the conversion of hard wax, which has a congealing point around 100 °C and consists of mostly *n*-paraffins in the C_{20}−C_{120} range. Unfortunately, the kinetic descriptions of Bragg[20] and Voge and Good,[21] as well as the Rice−Kossiakoff[22] description of the product distribution, were all developed using material no heavier than C_{40}. The first part of this study evaluated the applicability of these descriptions to thermal cracking of hard wax in order to propose a model that is valid for *n*-paraffins up to C_{120}. This could then be used to interpret the results obtained in the second part of this investigation, which evaluated the benefit of thermal cracking of hard wax to produce either fuels or chemicals.

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Experimental Section

Materials. All work was done with Sasol H8 paraffin wax (Table 1) from the commercial Arge LTFT synthesis at Sasol 1 in Sasolburg, South Africa. Sasol H8 is a hard wax that is obtained from vacuum distillation of the LTFT product. It has properties similar to those of Sasol H1 wax,[23] but has not been hydrogenated to remove the trace impurities responsible for the poorer color of H8 compared to that of H1 wax (Saybolt color +30). The hydrogen, nitrogen, and water (steam condensate) used as stripping gases were obtained from the Sasol 1 utility network.

Equipment and Procedure. Two reactors were used to study thermal cracking. The investigations using a stripping gas (hydrogen, nitrogen, or water) were performed in a fixed-bed reactor (1430 mm in length, with 28 mm internal diameter) in down-flow mode of operation. The reactor was filled with carborundum (0.7 mm diameter) to improve heat transfer and flow distribution. The top part of the reactor was used as a preheating zone and was filled with glass balls (4 mm diameter). Three independently controlled electric heaters maintained the temperature in the reactor at near-isothermal conditions. The investigations done without a stripping gas were performed in a 5 m long 1/8-in. stainless steel tube (1.9 mm internal diameter) that was coiled to fit inside the aforementioned reactor, which effectively served as an oven. This enabled operation at higher temperatures and shorter residence time. The operating range investigated was 420−500 °C, 0−6 MPa, and 0.1−1 h residence time. The possible contribution of conversion in the preheating zone to overall thermal cracking was evaluated, and no noticeable change was found in the wax properties at temperatures below 400 °C after 1 h residence time.

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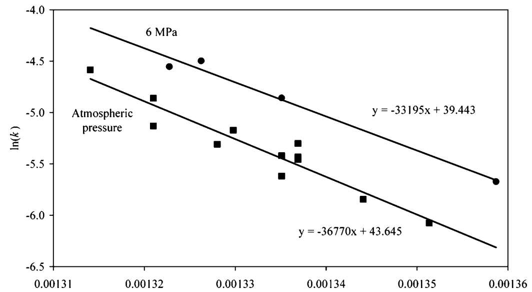

Figure 1. Van't Hoff plot of thermal cracking of Fischer–Tropsch wax in a continuous flow reactor, 1.9 mm internal diameter and 5 m long, at atmospheric pressure (■) and 6 MPa (●).

Table 1. Properties of Sasol H8 Paraffin Wax Obtained from Arge LTFT Synthesis

description	value
carbon number range	C_{20}–C_{120}
average molecular formula	$C_{50}H_{102}$
congealing point (°C)	98.5
penetration at 25 °C (mm)	0.1
MEK solubles (mass %)	<0.1
Saybolt color	0
distillation range (mass %)	
<370 °C	0.2
370–500 °C	4.1
>500 °C	95.7

Analyses. Physical characterization of the wax was done with the appropriate ASTM methods for paraffin waxes, for example, Saybolt color (D156), congealing point (D938), and penetration (D1321). The olefin distribution was determined by infrared spectrometry, and the metals content was determined by atomic absorption spectrometry.

Results and Discussion

Description of the Thermal Cracking of C_{20}-C_{120} Wax. A. Cracking Rate. Historically wax cracking processes were classified as gas phase or mixed phase. Gas-phase processes typically operated at a temperature of around 620 °C and low pressure, while mixed-phase processes operated in the temperature range 450–540 °C.[17] There are four parameters potentially affecting the thermal cracking of wax, namely temperature, pressure, residence time, and average carbon chain length of the material. Rate equations for the thermal cracking of *n*-paraffins and waxes in the temperature range 420–625 °C were previously reported, indicating an increase in cracking rate with temperature, as would be expected.[20,24,25] At lower temperatures (320–400 °C) thermal oligomerization of the olefinic products formed during cracking becomes significant.[26] In the liquid phase, cracking was found to be unimolecular and not influenced by pressure (eq 1).[20,24]

$$k = A \exp\left(\frac{-E_a}{RT}\right) = \frac{1}{t} \ln\left(\frac{1}{1-x}\right) \tag{1}$$

At lower pressures, typically less than 1 MPa, the cracking rate decreased with decreasing pressure and a slightly higher reaction order (1.3) was found to describe the data better.[25]

The thermal cracking of Sasol H8 hard wax was investigated experimentally in the range 460–490 °C at atmospheric pressure and at 6 MPa (Figure 1). First-order kinetics with a rate expression similar to eq 1 was used to analyze the results. The calculated apparent activation energies for thermal cracking of wax at atmospheric pressure and 6 MPa were calculated from the experimental data and found to be 305 ± 30 and 275 ± 25 kJ·mol^{-1}, respectively. These values are close to the value of 270 kJ·mol^{-1} reported for high-pressure wax cracking.[20] This may be explained as the sum of half the energy of the initiation process ($D_0 \approx 345$ kJ·mol^{-1})[27] and the energy of β-decomposition of the long chain alkyl radical (Figure 2).

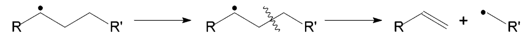

Figure 2. Thermal cracking of an *n*-paraffin radical, showing C–C bond rupture at the β-position from the hydrogen-deficient carbon to produce another radical and an α-olefin.

Considering the uncertainty in the calculated data, there is not a statistically meaningful difference between the apparent activation energies at different pressures. The Bragg[20] description of the wax cracking rate (eq 1) is therefore considered adequate for describing thermal cracking of waxes in the C_{20}–C_{120} range.

The thermal cracking rate constant also depends on the nature of the material being cracked. The cracking rate at a fixed temperature increases with increasing carbon chain length of the molecule being cracked,[21,25,28] and is described by an empirical correlation developed by Voge and Good[21] (eq 2).

$$k = (n-1)(1.57n - 3.9) \times 10^{-5} \tag{2}$$

However, Voge and Good[21] did not consider their empirical correlation (eq 2) to be a true relationship for all linear paraffins, but of limited validity for paraffin cracking, with $4 \leq n \leq 16$, at 500 °C and atmospheric pressure. The usefulness of this description for heavier paraffins will be considered with the description of the carbon number distribution of the product.

B. Carbon Number Distribution. The Rice–Kossiakoff description[22] of thermal cracking and the initial product distribution obtained by thermal cracking, still forms the basis for the more sophisticated mathematical descriptions that take the effect of secondary reactions into account.[29,30] According to their description, a pool of small alkyl radicals will form due to thermal cracking, eventually reaching a steady-state concentration. These alkyl radicals will remove hydrogen atoms from the wax molecules, producing a large radical that will rapidly decompose by C–C bond rupture at the β-position to from a hydrogen-deficient carbon (Figure 2). This in turn produces another radical and an α-olefin. Deviations from the predicted product distribution based on the Rice–Kossiakoff description becomes more significant as the cracking severity is increased and secondary reactions become more prominent. At temperatures above 650 °C the thermal cracking reactions reach equilibrium after about 5 s[31] and the product distribution is determined by Gibbs free energy minimization.

In the present study the experiments were done at mild conditions at incomplete wax conversion and far from equilibrium. The Rice–Kossiakoff description is therefore applicable to the product distribution. The carbon number distribution of the product shifted toward lighter products as cracking progressed, with a bimodal distribution developing at intermediate conversion (Figure 3). The cracking rate keeps increasing with increasing carbon chain length, at least until C_{90} and likely to C_{120}. (Accurate analysis of wax fractions in the C_{90}–C_{120} range is difficult.) Nevertheless, the data in Figure 3 suggest that the Voge and Good[21] description of increased cracking rate with increasing carbon chain length, as well as the Rice–Kossiakoff[22] description of the product distribution, may hold true for *n*-paraffin waxes in the C_{20}–C_{120} range.

C. Modeling. A model was developed based on the Voge and Good and Rice–Kossiakoff descriptions to describe thermal

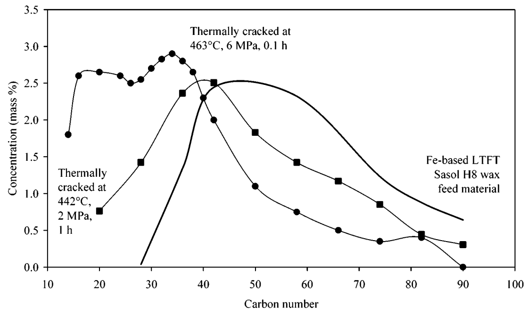

Figure 3. Carbon number distribution of the wax feed and products produced by thermal cracking at 442 °C, 2 MPa and 1 h residence time (■), and at 463 °C, 6 MPa and 0.1 h residence time (●).

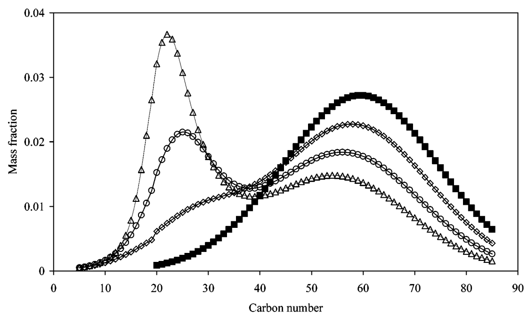

Figure 4. Carbon number distributions of the products from thermal cracking of a model wax feed as calculated by applying eqs 3 and 4 with $kB = 0.4 \times 10^{-5}$. The carbon number distributions at different levels of conversion of the $C_{50}+$ material are shown: 0% (■); 20% (◇); 40% (○); 55% (△).

cracking of wax. The carbon number dependent rate constant, $k(n)$, is calculated using eq 3.

$$k(n) = kB(n - 1)(1.57n - 3.9) \tag{3}$$

This expression is based on the Bragg reaction rate constant (eq 1) modified by the Voge and Good correlation (eq 2) to account for the influence of carbon number. The change in carbon number distribution with time is expressed by eq 4, using the Rice−Kossiakoff description as a basis.

$$\frac{dC(n)}{dt} = -k(n)\,C(n) + \left(\sum_{n=3}^{N} k(n)\,C(n)\right)\left(\frac{\exp\left(\frac{-(n - 0.5X(t))^2}{2s(t)}\right)}{\int_1^N \exp\left(\frac{-n^2}{2}\right) dn}\right) \tag{4}$$

The first term in eq 4 is the first-order thermal cracking rate to consume the species, while the second term gives the production rate of the species as a function of the total cracking rate and the probability that this species will be formed. The underlying assumption used for the second term is that the radical products will have a normal distribution around a mean that is half the average chain length. The time-dependent average chain length, X, and standard deviation, s, determine the product distribution.

When eqs 3 and 4 are used to calculate the time (or conversion) dependent carbon number distribution (Figure 4), the development of a bimodal distribution and a noticeable shift in carbon number distribution of the heavy fraction are observed. From a comparison of Figures 3 and 4 it can be seen that the model is capable of describing thermal cracking qualitatively. Since the aim of the model was to aid interpretation of the experimental data, rather than predicting conversion, the constants implicit in eq 3 were not optimized. Should a quantitative description be required, the constants will have to be determined by data regression.

Application of the Thermal Cracking of C_{20}-C_{120} Wax. A. Fuels Refining. One advantage of thermal cracking for fuels production, which is apparent from eq 3, is that the cracking rate decreases with decreasing carbon chain length. It thereby suppresses the overcracking of naphtha and distillate to lighter products and explains why thermal cracking inherently has a low gas make. The gas make can be further reduced by removing the lighter products from the liquid wax either by making use of the gas phase being formed during cracking, or by co-feeding a stripping gas. For the former to be effective, the operating pressure should be sufficiently low for a gas phase to form, although it was shown that a gas phase develops during cracking even at operating pressures above 3 MPa.[20] By co-feeding a stripping gas it is possible to operate under elevated pressures and remove light products even at low conversion, resulting in little gas make (Table 2).

When thermal cracking is compared to hydrocracking for the production of <370 °C boiling material from wax (Table 3), it is clear that hydrocracking is more efficient. At similar conversion (20−30%) and space velocity (LHSV of 1 h^{-1}), a hydrocracker can be operated about 70 °C lower than a thermal cracker. Furthermore, at their respective operating temperatures, hydrocracking conversion increases faster with an increase in temperature than thermal cracking. For example, using the apparent activation energies for the hydrocracking of long-chain *n*-paraffins (230−265 kJ·mol^{-1}),[32] the calculated average increase in hydrocracking conversion when the temperature is increased from 370 to 380 °C is 103%. The calculated increase in thermal cracking when the temperature is increased from 440 to 450 °C using the apparent activation energy of Bragg (270 kJ·mol^{-1})[20] is only 88%.

If diesel fuel is the main aim of the refining operation, it has been reported that at 360 °C, 3.5 MPa, and liquid hourly space velocity of 0.55 h^{-1} a distillate yield of 63% can be obtained by LTFT wax hydrocracking.[5] Similar results have been reported by Shell.[3,4] The use of thermal cracking for the production of transportation fuels was not investigated further, since little advantage over hydrocracking could be shown.

B. Chemicals Refining. It is less important to convert wax into material that boils in the <370 °C range when the aim is to produce chemical products. The high olefin content of the thermally cracked products gives them synthetic value, especially since it is rich in linear α-olefins (Table 2). The partially converted wax can also find application in wax-based products, with the medium wax cut (C_{19}−C_{38}) having a large market as candle wax.[23]

There are two approaches that can be followed to make use of thermal cracking for the production of candle wax from hard wax: (a) thermal cracking followed by topping and tailing to separate the medium wax from the product, and (b) thermal cracking followed only by topping to remove the naphtha and distillate fraction. Although the first option is similar to the primary separation of the LTFT product at the Sasol 1 refinery,[33] it requires high vacuum distillation to separate the medium wax from the hard wax. Despite the additional capital and operating cost, it has the advantage that the wax used for cracking becomes heavier, which according to the thermal cracking model (eqs 3 and 4) would yield mainly candle wax as product during moderate thermal conversion (30−40% per pass). The second

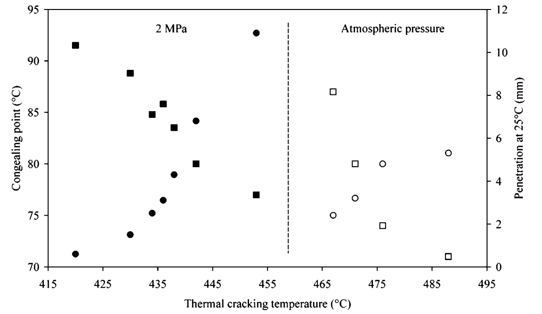

Figure 5. Congealing point (■) and penetration (●) of Sasol H8 wax after thermal cracking at 2 MPa, 1 h residence time, and hydrogen at 250 $m^3 \cdot m^{-3}$ wax flow (solid symbols), and atmospheric pressure, 0.1 h residence time, and no stripping gas (open symbols).

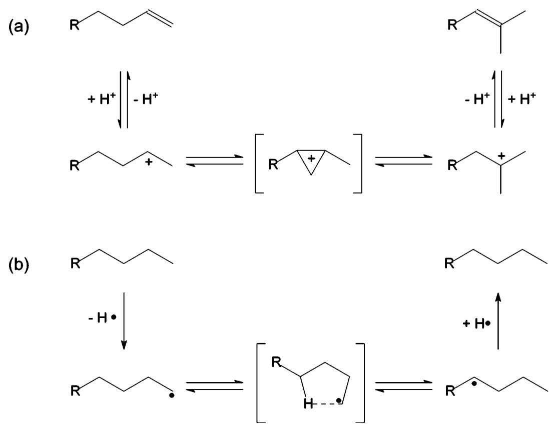

Figure 6. Isomerization during (a) acid-catalyzed reactions via a carbocation intermediate that leads to skeletal rearrangement and (b) radical reactions via 1,4 or 1,5 (not shown) hydrogen transfer, which does not lead to skeletal rearrangement.

option, which uses only atmospheric distillation to remove the light-boiling material, is an attractive processing option from a cost point of view, but will result in higher naphtha and distillate production at similar conversion as the first option. A more detailed economic analysis is required to select the most profitable option, and in the present study only the second option is further explored.

If the thermally cracked product is to be used as candle wax, it must conform to the properties needed to make good candles. Ideally the wax should have a congealing point of around 55−60 °C (to avoid burn wounds when hot candle wax is dripped onto the skin), penetration of 3−4 mm at 25 °C (to avoid excessive breakage when dropped), and less than 2% material heavier than C_{50} (to avoid guttering of the flame because the high-viscosity heavy wax restricts wax flow to the wick).

When the wax was thermally cracked at mild conditions (Figure 5), it became clear that the penetration and congealing point specifications would not be simultaneously met. The penetration, which is related to the degree of isomerization, was strongly influenced by pressure. Unlike acid-catalyzed cracking, where skeletal isomerization is possible by intramolecular rearrangement of the carbocation intermediate, thermal cracking produces radical intermediates that isomerize by 1−4 and 1−5 hydrogen transfer[29,34] and does not lead to skeletal isomerization (Figure 6). However, at high pressure, radical addition to olefins is promoted. This introduces branching into the molecular structure when the radical attack occurs on a nonprimary carbon atom (eq 5), but not when the attack occurs on a primary carbon atom.

$$RCH{=}CH_2 + \grave{R}^{\bullet} \rightarrow RC\grave{R}H{-}CH_2^{\bullet} \qquad (5)$$

Changing the thermal cracking conditions to lower pressure, while increasing the temperature and reducing the residence time, nevertheless drove product quality in the right direction (Figure 5), but the target was still not met. The congealing point was related to the heavier than C_{50} fraction, and for thermal cracking at atmospheric pressure it is described by eq 6. The heavier than C_{50} target would therefore be met when the congealing point target is met.

$$>C_{50}\ \text{fraction} = 0.6(\text{congealing point, °C}) - 34 \qquad (6)$$

We have not been able to find thermal cracking conditions where the key candle wax properties could be achieved without topping and tailing the cracked wax. Nevertheless, it was possible to blend thermally cracked wax (488 °C, 0 MPa, 0.15 h residence time) with straight-run LTFT medium wax (Arge M5) in a 1:1 ratio to obtain a candle wax that meets the physical property specifications. The wax color, which has not been considered during this investigation, can be corrected by hydrogenation.[35]

C. Deposit Formation. Deposit formation may result in operational difficulties during the thermal cracking of Fischer−Tropsch wax. The problems associated with such deposits are

Table 2. Product Distribution Obtained during Thermal Cracking of Wax at 2 MPa and 1 h Residence Time with Hydrogen at 250 $m^3 \cdot m^{-3}$ Wax as Stripping Gas

description	product distribution (mass %)		
	434 °C	438 °C	442 °C
gas (C_1−C_4)	2	2	2
naphtha and distillate (C_5-370 °C)	13	16	20
vacuum distillate (370−500 °C)	24	24	26
vacuum residue (>500 °C)	61	58	52
olefins in C_5-370 °C fraction		48	42
α-olefins in C_5-370 °C fraction		41	39

Table 3. Comparison of Thermal Cracking and Hydrocracking (Sulfided Ni/Mo−SiO_2/Al_2O_3 Catalyst)[5] of Arge LTFT Wax at Similar Conversion

description	thermal cracking with stripping gas	thermal cracking	hydrocracking
temperature (°C)	442	476	370
pressure (MPa)	2	0	3.5
liquid hourly space velocity (h^{-1})	1	8	1
hydrogen flow (normal $m^3 \cdot m^{-3}$ wax)	250	0	1500
yield of C_1−C_4 (mass %)	2	1	3
yield of C_5-370 °C (mass %)	20	21	28
naphtha (C_5−C_9):distillate (C_{10}−C_{22})		1:9	1:7

Table 4. Metals Present in the Wax before and after Thermal Cracking, Which are Typically Associated with the Iron-Based LTFT Catalyst That Was Used for Wax Synthesis

metal	wax feed ($\mu g \cdot g^{-1}$)	cracked wax ($\mu g \cdot g^{-1}$)
Na	0.5	0.4
K	0.2	0.1
Fe	4.3	1.8
Cu	<0.1	<0.1

accentuated during laboratory studies, where reactors can easily be blocked on account of their small diameters.

Previous work on the thermal oligomerization of Fischer−Tropsch olefins did not result in bed plugging during 78 days of continuous operation, although some iron-rich (40% Fe) deposits were formed.[26] In comparison, deposit formation during thermal cracking of Fischer−Tropsch wax was rapid, with bed plugging being found in less than 18 days. Failure to unplug the reactor by controlled carbon burnoff necessitated physical removal of the deposits, which had a high inorganic content. Analysis of the H8 wax feed revealed silica (60 $\mu g \cdot g^{-1}$), iron (4 $\mu g \cdot g^{-1}$), and nickel (3 $\mu g \cdot g^{-1}$) as the main inorganic constituents of the wax, which were also found in the deposits. It is not clear to what extent these deposits contributed to the cracking reactions and further deposit formation once they were formed.

There are two important sources of inorganic impurities in Fischer−Tropsch derived material, namely corrosion and catalyst derived products. The carboxylic acids produced during Fischer−Tropsch synthesis are able to cause acid leaching of metals, especially at the temperatures that are typically found during synthesis and postprocessing. Analysis of the wax before and after thermal cracking (Table 4) showed that only some of the metals present in the precipitated iron-based LTFT Arge fixed-bed catalyst[36] being used at Sasol 1 for wax production were also found in the wax. The iron species, most likely iron carboxylates, were thermally the most labile and decomposed at the operating conditions used for wax cracking (Table 4) to yield the iron-rich deposits that were found experimentally. Others have also reported the presence of iron in Fischer−Tropsch derived feed materials and the deposition of iron species during the refining thereof.[37−39]

Carbonaceous deposit formation was much lower than suggested by the plugging propensity of the feed. It was found to be 70 $\mu g \cdot g^{-1}$ for operation that involved ramping of the temperature from 260 to 442 °C and 30 $\mu g \cdot g^{-1}$ for isothermal operation at 434 °C. At lower temperatures, typically in the range 320−400 °C, thermal oligomerization can produce heavy oligomers that contribute to the formation of carbonaceous deposits,[26] while at higher temperatures thermal cracking reduces the contribution of such deposits.

Since the deposits were rich in inorganic matter, it was doubtful whether diluents would be beneficial in reducing deposit formation. In commercial thermal cracking operations, which are operated at temperatures around 800 °C, steam is used to reduce the hydrocarbon partial pressure. The steam increases selectivity to desired products and reduces unwanted side reactions like coke formation.[40,41] However, steam was found to be ineffective in preventing the formation of deposits during thermal cracking of wax. Operation at 465−485 °C resulted in plugging after 18 days of continuous operation, while operation at 510 °C resulted in plugging within 4−5 days. Additionally, the product contained less olefins and 0.1−0.2% alcohols. The alcohols may have been formed by the reaction of steam with the wax, but the water was not deaerated and the alcohols could also have been formed by autoxidation.[42] The use of hydrogen and nitrogen were similarly ineffective in reducing deposit formation. The product obtained with hydrogen as stripping gas was more hydrogenated (improvement in Saybolt color) and branching was less due to the lower concentration of olefins (eq 5), resulting in a higher congealing point.

Conclusions

The present work investigated the thermal cracking of Fischer−Tropsch hard wax and focused mainly on operation in the lower temperature range (420−500 °C) of mixed-phase cracking. Two aspects were considered, namely, the description of thermal cracking of C_{20}−C_{120} wax and the application of thermal cracking to convert wax into fuels and chemicals. The following were found:

(a) The description of *n*-paraffin cracking recorded in the literature was mainly developed using material no heavier than C_{40}. The descriptions of Bragg,[20] Voge and Good,[21] and Rice−Kossiakoff[22] of the product distribution were extended to C_{20}−C_{120} wax. The relationship between carbon number distribution and conversion could be modeled qualitatively and indicated that cracking rate increased with increasing carbon length, with thermal cracking resulting in a bimodal product distribution.

(b) An apparent activation energy of 305 ± 30 kJ·mol^{-1} was obtained for thermal cracking of wax at atmospheric pressure. At high pressure the calculated apparent activation energy was 275 ± 25 kJ·mol^{-1}, which is close to the value of 270 kJ·mol^{-1} previously reported by Bragg.[20]

(c) In comparison to hydrocracking, it was found that thermal cracking is less efficient for the production of diesel fuel and that it is better suited for chemicals production. The <370 °C boiling fraction has a high linear α-olefin content (about 40%) and the medium wax fraction can find application as candle wax.

(d) The production of candle wax was explored using a process based on thermal cracking of Fischer−Tropsch wax (C_{20}−C_{120}) using only atmospheric distillation to remove the lighter fraction. It has been found that with this process the candle wax specifications could only be met when the product was blended with straight-run medium wax (C_{19}−C_{38}).

(e) Deposits that were formed during the thermal cracking of Fischer−Tropsch wax were rich in iron, and their formation could not effectively be reduced by the use of stripping gases (steam, hydrogen, and nitrogen).

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Nomenclature

A = preexponential frequency factor (s^{-1})
B = correlation constant to relate k to $k(n)$ in eq 3
$C(n)$ = mass fraction of species with chain length n
D_0 = homolytic bond dissociation energy (kJ·mol^{-1})
E_a = activation energy (J·mol^{-1})
k = first-order rate constant (s^{-1})
$k(n)$ = first-order rate constant for species with chain length n (s^{-1})
n = number of carbon atoms per molecule
N = chain length of the molecule with the largest n

R = universal gas constant, 8.314 J·mol^{-1}·K^{-1}
s = standard deviation of carbon number distribution
t = residence time in reactor (s)
T = temperature (K)
x = fraction of C_{50} and heavier wax converted
X = average carbon number

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